

MANIBHARATHI A

Entry-Level Data Analyst | Python, SQL, Tableau | Skilled in ML & Data Visualization |
8838154112 | amanibharathi2006@gmail.com | **Portfolio:** <https://manibharathi2006.netlify.app/>

SUMMARY

Computer Science student and aspiring AI Engineer specializing in Generative AI and Computer Vision. Experienced in building RAG pipelines and real-time object detection systems. Focused on developing end-to-end full-stack AI applications with a minimalist and professional approach.

SKILLS:

- **Generative AI:** RAG, LangChain, LLM APIs (OpenRouter), Embeddings.
- **Computer Vision:** YOLOv8 Object Detection, Image Preprocessing, Real-time Monitoring.
- **Databases:** FAISS, ChromaDB, Qdrant, MySQL, MongoDB, Firebase.
- **Programming:** Python, Java (Proficient), SQL, Node.js, React, HTML/CSS/JS.
- **Tools:** Git, Tableau, Excel, Pandas, NumPy.

Experience:

- **GenAI Intern | Shellkode Dec 2025 – Jan 2026**
 - Developed a RAG pipeline using LangChain to enable AI-powered querying of private documents.
 - Optimized document ingestion and chunking strategies for FAISS and Qdrant vector stores.
 - Implemented semantic search to improve retrieval accuracy for LLM-generated responses.
- **Data Science Intern | Shen AI Software Solutions June 2025 – July 2025**
 - Built a YOLOv8 fire detection model with 80%+ confidence for real-time video feeds.
 - Integrated Telegram Bot API and Geocoder for automated alerts with live GPS locations.

PROJECTS:

- **TypeHype: RAG-Based Document Intelligence**
 - Engineered a full-stack RAG system utilizing Llama 3 and Groq API for context-aware document interaction.
 - Managed the end-to-end workflow from PDF ingestion and recursive character text splitting to semantic retrieval.
 - Leveraged ChromaDB to store word embeddings, ensuring efficient and precise similarity searching.
- **Wild Animal Detection & Repellent System**
 - Created an AI-IoT system using YOLOv8 to detect wildlife and prevent human-wildlife conflict in real-time.
 - Integrated Twilio API for automated voice calls and Telegram for instant evidence sharing.
 - Designed a hardware trigger for an ultrasonic repellent to deter animals non-invasively.
- **Fire Detection & Alert System**
 - Developed a computer vision safety tool for smart homes and industrial monitoring with live video analysis.
 - Programmed the system to automatically capture incident images and log timestamps for audit trails.

EDUCATION

Nandha Engineering College (Autonomous)
B.E - Computer Science and Engineering | 2023 - 2027 | Tamil Nadu CGPA - 8.4

ONLINE PROFILES

GitHub Profile LinkedIn Profile

ACHIEVEMENTS

- **Hackathon – 2nd Prize (Data Analytics)**
 - Developed dashboards using Tableau and Excel to gather useful insights.
 - Created data storytelling solutions that highlighted project impact. LeetCode
- **DSA Practice**

Solved over 149 problems using Java and SQL.